Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 5

Part A

tetrahedral because there are 4 electron domains around Si

Part B

Yes, Si has more electron shells than C, so it has a more polarizable electron cloud than C, so compound Y must be more polarizable than compound X, so compound Y will have greater london dispersion forces and a higher boiling point than compound X.

Part C

Compound X will have the higher vapor pressure because more of compound X will be in vapor form at 82°C than compound Y because compound X's boiling point is equal to 82°C and compound Y's boiling point is greater than 82°C (98°C).

Question 5 is continued on the next page.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 5

Part D

$PV = nRT$

$(2.30\ atm)(12.5L) = n\ (0.08206\ L\,atm\,K^{-1}\,mol^{-1})(471.15K)$

$\boxed{n = 0.744\ moles}$

Continue to Question 6.